

This is not a classroom: Theorizing affordances for learning

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Figure 1: Virtual Reality Environment designed as College Classroom, photo credit McEwan Mediated Communication Lab .

ABSTRACT

Virtual Reality provides many opportunities to move beyond the bounds of physical classrooms in the pursuit of designing effective learning environments. A key design consideration of VR learning environments are the potential perceived affordances that might influence learning outcomes. However, design may influence different types of learning in diverse ways. This position paper outlines the importance of affordances, presents the case of perceived environmental fidelity and proposes a set of additional affordances that designers, educators, and researchers should consider.

Index Terms: Virtual Reality, Affordances, Environmental Fidelity, Immersion, Presence, Persistence, Learning

1 INTRODUCTION

Virtual reality (VR) and immersive spaces have incredible potential for both distance learning applications as well as providing novel and exciting experiences within classroom environments. Within this position paper we posit that educators should undertake a careful assessment of the "mosaic of technologies" [22] that make up virtual reality. Further, it is important to consider how virtual reality can facilitate, enhance or in some cases hinder specific learning outcomes. For future research in this area we take the position that it may be more important to evaluate how affordances of particular

VR programs influence educational impacts than to compare VR settings to other learning sites.

2 VIRTUAL REALITY AFFORDANCES

Virtual reality, or immersive virtual environments that occlude the physical world and are perceived via the user's sensorimotor input [20] is a potentially powerful tool for education. However, the design of VR software and hardware and principles of human cognition and communication intersect in ways that can make producing and evaluating educational outcomes challenging. Yet, in-depth understanding of these intertwined elements could improve the educational process.

To understand the utility of VR for cognition and education, it is useful to draw lessons from affordances of mediated channels. Affordances are the "multidimensional relationship between the object or technology and the user and how that relationship offers possible (and actual) outcomes" [11]. Affordances exist on a continuum and while affordances are neither feature nor outcome, the possibilities that make up an affordance are grounded within the features of a technology, the perception of the users, and the social context and constructions of a technology as they are understood by the user [11, 24]. Considerations of affordances in VR extend beyond the work on affordances of physical objects [26] and the work on perceived social affordances of communication channels [12]. Virtual environments combine object affordances (what can we do with this virtual block?), design affordances (how does the user interact with this block within this user interact with this block within this software through the hardware controls?), and social affordances (does the way I interact with the block signal some message to co-present others?). Understanding perceived affordances can help educators and researchers match learning goals with the affordances of social VR spaces [5, 10].

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Starting from the perspective of affordances can engage scholars in layered and fruitful research questions investigating the different potential effects of these affordances. One important question is whether, given that affordances rely partially on context, is if the affordances of social VR spaces are tied to heuristics developed in the physical world. Thus, the case of examining physical fidelity as an affordance related to education may provide a useful starting point for considering a wide range of potential affordances and their connection to academic possibilities.

3 A CASE STUDY: FIDELITY

VR for education has the opportunity to introduce students to fantastical and imaginative worlds. Given this, the question arises of why so many VR companies in the education space (e.g. ENGAGE, VirBela, Frame) design standard lecture halls, when one of the potential VR affordances is the creation of environments that can have considerably less fidelity to the physical world. We posit that perceived environmental fidelity may have different effects on different types of learning activities.

Previous research has found the type of virtual environment may influence learner outcomes. For example, using the ENGAGE platform, Han et al. [14] found that the type of environment has an effect on users in a VR course. As visible space increased communication variables such as nonverbal synchrony, entitativity (how connected to the group one feels), and presence increased. In addition, outdoor environments were perceived as more enjoyable than indoor environments. This work suggests that virtual “classrooms” should perhaps not be constrained to match physical classrooms. Environments that do not resemble physical classrooms could have a greater impact on collaboration and learning.

At the same time, VR work in academia and industry has also recently focused on the concept of “Digital Twins;” virtual objects that mirror a real-world counterpart. While the idea of digital twins originated from creating a digital version of a product, digital twinning has also influence the creation of training environments. In this use case, a virtual environment is created to replicate the state of a physical environment for the purpose of training people within that environment [16]. Underpinning the idea of digital twin environments is that users will be able to develop cognitive heuristics within the virtual environment that can be applied to the physical world [35]. However, more research is needed to determine the success of such training environments as well as the ways in which cognitive heuristics developed in the physical world influence perceptions, behaviour, and cognitive load within the virtual world. Varying levels of environmental fidelity may have different outcomes for learning [10]. On one hand, following from work in psychology [8, 23] and educational technology [19, 29] we might expect that environments that do not easily fit within existing heuristics of social spaces create additional stresses and cognitive load for users. For individuals working within VR, a common issue reported by researchers is that cognitive overload can hinder learning. For example, in a study that examines cognitive load directly through EEG measures, 52 university students were offered either a desktop or VR simulation (via head-mounted displays) of a biology laboratory activity [19]. The study showed that participants who engaged with the desktop version had significantly higher knowledge gain than those in the immersive VR group. This was accompanied by higher average workload obtained from EEG data, which supports prior concerns regarding cognitive overload in immersive VR. Cognitive load results from the short-term information processing activities of the mind when various information elements are being held and manipulated simultaneously [21]. The short-term system for storing and manipulating information is called working memory [1]. Within educational psychology, cognitive load theory is used to examine difficulties that arise in learning scenarios and can be used to identify opportunities for manipulating content to reduce

and/or improve information processing [32]. It is most useful when the learning content is amenable to manipulation by instructional design. We can use techniques and practices in instructional design to reduce information processing by examining ways to target the efficient use of working memory in learners. Working memory is the crucial part of processing where information is held for a limited amount of time before passing into long-term memory before fading away [31]. Virtual environments with high physical fidelity may allow for the activation of existing cognitive heuristics and greater access to working memory. Whereas more imaginative environments may have greater novelty and initial interest but could restrict learners’ available working memory.

However, while the effects of virtual environments may have a deleterious effect on learning tasks related to working memory, the de-yoking of environment from the constraints of physical reality could allow for the possibility of breaking out of cognitive heuristics and social schemas and allow for greater creativity of thought and output. The standard definition of creativity is the ability to combine originality and effectiveness [28]. For a product, an idea, or a solution to be viewed as creative, it must be innovative, original, and perhaps even surprising, but it also must be appropriate for the context requiring the idea [25, 27]. Creativity goes beyond a simple “aha” moments and is related to cognitive processes associated with defining problems, gathering knowledge, combining concepts, generating ideas, and implementing those ideas [23, 33].

Although some people may have a predisposition to be more creative [2], there is also evidence that creativity can be fostered through environmental influences [7, 30]. We argue that one possible opportunity for virtual environments is that low fidelity environments might encourage higher levels of creativity by freeing users’ minds from the constraints of the physical world. Bruner argued that those wishing to be creative must first break away from the obvious in order find the surprising and fresh combinations that are perceived to be creative. More recent creativity researchers (e.g., [3]) have posited that designing learning activities in an unexpected way might foster creativity. We wonder if unexpected virtual environments unbound from the constraints of physical reality might also have a positive impact on creativity. Specifically, video game research has found that games with more undirected creativity may lead to engaging in more creative thinking [4, 15]. Thus, the level of perceived environmental fidelity may influence the type of learning that can occur within in VR.

4 AFFORDANCES OF PEDAGOGICAL INTEREST

While our current program of research is investigating the effects of perceived environmental fidelity on diverse learning outcomes, we argue that other potential affordances are also of interest for scholars of VR and education. These include of course immersion, but also social presence, bandwidth, and persistence.

Immersion, or how encapsulated a user feels in a virtual world [6, 13], is quite likely the most often considered affordance for why virtual reality might be useful for educational settings. Immersion may indeed be useful for focusing student attention on the educational content. However, immersion may also distract the student and increase cognitive load. The proliferation of mixed reality tools also will require educational technologists to consider whether it is truly immersion that predicts the effects of digital educational tools, as these educational settings are digital but not immersive.

In their Cognitive Affective Model of Immersive Learning (CAMIL) Makransky and Petersen [18] posed that agency may be an important affordance. While the term agency seems closer to a user attribute, the idea that they convey with it – a user’s perception of how malleable the environment is to the user’s specific whims is an affordance. VR spaces that allow for interaction and editing of objects can provide different pedagogical choices than immalleable programs. Another aspect of malleability is the ability to change

the respective size of objects and students. This could allow for engaging inspections of tiny atoms and immense solar systems [35].

Social presence, or the feeling of being ‘there’ with other people [17], has also been oft argued, including in CAMIL, as an important affordance for VR adoption [18]. The ability to feel “there” with others despite geographic distances can allow for collaborative educational opportunities. However, this sense of social presence can also be met through other technological tools such as zoom [9]. The avatar-to-avatar communication common in virtual environments may offer less bandwidth, (i.e. the nonverbal cues that support social and emotional communication, see [12]), that the video conferencing (although the avatars may have *more* bandwidth than students in zoom black boxes).

Bandwidth itself may be an important affordance for pedagogy. The practice of lecturing can feel awkward and challenging when there is a lack of nonverbal cues between the speaker and the audience. In addition, the nonverbal cues that bandwidth affords are an important part of the relational-level meaning [34] that is important for many different types of classroom interactions including class discussion and small group interactions. Different VR environments may have differing options for conveying nonverbal information that would need to be assessed by the educator.

Another important affordance might be persistence. Although physical face-to-face environments are often considered just as ephemeral as communication within virtual environments, in physical classrooms students often have access to other technologies (e.g. pen and paper, word processors, digital cameras) that allow the messages being conveyed between the student and educator to have some level of persistence. Note-taking in virtual environments is currently often an afterthought or completely non-present in some designs. Recordings may be possible, but design features that enhance persistence may allow for greater review and retention by students.

5 CONCLUSION

Research on defining and measuring the relevant affordances in VR for educational purposes is still at a nascent stage. The affordances discussed herein are important ones for the consideration of educational technologists, teachers, and learners but they are certainly not the only ones. Research should continue to investigate the relationship between design, affordances, and potential educational outcomes. Educators should consider these affordances and their theoretical alignment with their pedagogical goals but also remember that a considerable amount of work has yet to be done explicating the specific effects of these affordances.

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